

Control work 2 — division of fractions, and work on the mistakes

The four operations on fractions in one paper, then the repair.

LESSON 36–37

2 × 40 MIN

MATEMATIKA 6, NAZORAT ISHI 2

S7 7 REVIEW

LESSON CLOCK

3

35'

12'

25'

5'

Instructions	3 min
The paper	35 min
Answers	12 min
Diagnosis and rewrite	25 min
The map	5 min

LEARNING OBJECTIVES

- Work accurately with all four operations on fractions.
- Handle mixed numbers correctly.
- Solve a word problem, naming the operation.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 54–98
Cambridge	Stage 7, pp. 66–80
Workbook	Control paper 2

01

Explanation

The paper — 25 marks, 35 minutes

Q	TASK	MARKS	FROM
1	$\frac{3}{4} + \frac{2}{5}$ and $\frac{7}{8} - \frac{1}{4}$	4	L21
2	$\frac{4}{9} \cdot \frac{3}{8}$	3	L21
3	$\frac{5}{6} \div 5$ and $8 \div \frac{2}{3}$	4	L22–26
4	$\frac{2}{3} \div \frac{8}{9}$	4	L27–29
5	$2\frac{1}{4} \div 1\frac{1}{2}$	4	L27–29
6	$\frac{3}{5}$ of a tank is 24 litres: find its capacity	3	L30–32
7	How many $\frac{3}{4}$ m planks come from 9 m?	3	L30–32

THE ANSWERS

$1\frac{3}{20}$ and $\frac{5}{8}$; $\frac{1}{6}$; $\frac{1}{6}$ and 12; $\frac{3}{4}$; $1\frac{1}{2}$; 40 litres; 12 planks.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
denominators added	$\frac{3}{4} + \frac{2}{5} = \frac{5}{9}$	use the LCD 20
cancelled after multiplying	$\frac{12}{72}$	cancel first: $\frac{1}{6}$
numerator multiplied when dividing by a whole number	$\frac{25}{6}$	$\frac{1}{6}$
wrong fraction inverted	$\frac{3}{2} \cdot \frac{8}{9}$	$\frac{2}{3} \cdot \frac{9}{8}$
mixed number not converted	$2\frac{1}{4} \div 1\frac{1}{2} = 2\frac{1}{6}$	$\frac{9}{4} \div \frac{3}{2}$
multiplied instead of divided	$24 \cdot \frac{3}{5}$	$24 \div \frac{3}{5}$
answer not simplified	$\frac{9}{12}$	$\frac{3}{4}$

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

The chapter as one map

CASE	METHOD	ANSWER COMPARED WITH THE FIRST NUMBER
fraction $\div$ whole number	multiply the denominator	smaller
whole number $\div$ fraction	multiply by the reciprocal	larger
fraction $\div$ fraction under 1	multiply by the reciprocal	larger
fraction $\div$ fraction over 1	multiply by the reciprocal	smaller
mixed numbers	convert first, then as above	depends on the divisor

LOOKING FORWARD

Next come three short Cambridge lessons — common multiples, divisibility and roots — and then ratio, where dividing fractions returns as “how many times bigger is this than that”.

02 Worked examples

EXAMPLE 1 Model answer, Q3: $\frac{5}{6} \div 5$ and $8 \div \frac{2}{3}$.

$$\textcircled{1} \quad \frac{5}{6} \cdot \frac{1}{5}$$

Cancel the 5s.

$$\textcircled{2} \quad = \frac{1}{6}$$

Smaller ✓

$$\textcircled{3} \quad 8 \cdot \frac{3}{2}$$

Turn over the divisor.

$$\textcircled{4} \quad = 12$$

Larger ✓

ANSWER

$\frac{1}{6}$ and 12

EXAMPLE 2 Model answer, Q5: $2\frac{1}{4} \div 1\frac{1}{2}$.

① Convert: $\frac{9}{4} \div \frac{3}{2}$.

Never divide mixed numbers directly.

② $\frac{9}{4} \cdot \frac{2}{3}$

③ Cancel and multiply.

④ $= \frac{3}{2} = 1\frac{1}{2}$

ANSWER

$$1\frac{1}{2}$$

EXAMPLE 3 Model answer, Q6: $\frac{3}{5}$ of a tank is 24 litres.

① A part is given and the whole is wanted: divide.

② $24 \div \frac{3}{5} = 24 \cdot \frac{5}{3}$

③ $= 40$ litres.

Larger than 24 ✓

ANSWER

40 litres

03 Interactive model

Return Q6 and Q7 first and ask which operation each needed; almost every lost mark there is a choice of operation, not arithmetic.

The chapter in eight questions

3 4 + 2 5 equals:

$$\frac{5}{9}$$

$$1\frac{3}{20}$$

$$\frac{6}{20}$$

$$1\frac{1}{20}$$

4 9 · 3 8 equals:

$$\frac{1}{6}$$

$$\frac{12}{72} \text{ unsimplified}$$

$$\frac{7}{17}$$

$$\frac{32}{27}$$

5 6 ÷ 5 equals:

$$\frac{1}{6}$$

$$\frac{25}{6}$$

$$\frac{5}{30} \text{ unsimplified}$$

$$\frac{6}{5}$$

8 ÷ 2 3 equals:

$$\frac{16}{3}$$

$$12$$

$$\frac{4}{3}$$

$$5\frac{1}{3}$$

2 3 ÷ 8 9 equals:

$$\frac{16}{27}$$

$$\frac{3}{4}$$

$$\frac{4}{3}$$

$$\frac{27}{16}$$

$2\frac{1}{4} \div 1\frac{1}{2}$ equals:

$$1\frac{1}{2}$$

$$2\frac{1}{6}$$

$$3\frac{3}{8}$$

$$\frac{2}{3}$$

$\frac{3}{5}$ of a tank is 24 l; the tank holds:

$$14.4 \text{ l}$$

$$40 \text{ l}$$

$$30 \text{ l}$$

$$72 \text{ l}$$

9 m in $\frac{3}{4}$ m planks gives:

$$6.75$$

$$12$$

$$9$$

$$3$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Division of fractions	Kasrlarni bo'lish	Деление дробей
Reciprocal	Teskari son	Обратное число
Mixed number	Aralash son	Смешанное число
Common denominator	Umumiy maxraj	Общий знаменатель
Lowest terms	Qisqarmas ko'rinish	Несократимый вид
Word problem	Matnli masala	Текстовая задача
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 needs:

Q2 is quickest if you:

Q3 first part gives an answer:

Q3 second part gives an answer:

larger

smaller

equal

negative

Q5 must begin with:

turning over

converting the mixed numbers

a common denominator

cancelling

Q6 needs:

multiplication

division

addition

subtraction

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $\frac{3}{4} + \frac{2}{5}$

ANSWER $1\frac{3}{20}$

2. $\frac{7}{8} - \frac{1}{4}$

ANSWER $\frac{5}{8}$

3. $\frac{4}{9} \cdot \frac{3}{8}$

ANSWER $\frac{1}{6}$

4. $\frac{5}{6} \div 5$

ANSWER $\frac{1}{6}$

5. $8 \div \frac{2}{3}$

ANSWER 12

6. $\frac{2}{3} \div \frac{8}{9}$

ANSWER $\frac{3}{4}$

7. $2\frac{1}{4} \div 1\frac{1}{2}$

ANSWER $1\frac{1}{2}$

Medium · the standard question

7 PROBLEMS

1. $\frac{3}{5}$ of a tank is 24 l

ANSWER 40 l

2. 9 m in $\frac{3}{4}$ m planks

ANSWER 12

3. $1\frac{1}{2} + 2\frac{3}{4}$

ANSWER $4\frac{1}{4}$

4. $5 \div \frac{5}{8}$

ANSWER 8

5. $\frac{7}{10} \div \frac{7}{5}$

ANSWER $\frac{1}{2}$

6. $\frac{3}{8} \cdot 4$

ANSWER $1\frac{1}{2}$

7. $\frac{9}{10} \div 3$

ANSWER $\frac{3}{10}$

Hard · stretch and reason

7 PROBLEMS

1. $3\frac{1}{3} \div 2\frac{2}{9}$

ANSWER $1\frac{1}{2}$

2. $\frac{5}{6} + \frac{3}{4} \div \frac{3}{2}$

ANSWER $1\frac{1}{3}$

3. A tank $\frac{7}{8}$ full holds 35 l: its capacity

ANSWER 40 l

4. How many $\frac{2}{5}$ kg bags from 6 kg?

ANSWER 15

5. $1 \div \frac{3}{7}$

ANSWER $2\frac{1}{3}$

6. After using $\frac{3}{8}$, 30 l is left: the original

ANSWER 48 l

7. Why is the answer to Q3 second part larger than 8?

ANSWER The divisor is smaller than one

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on in full, from the first line.

1. Rewrite in full every question on which you lost a mark.

2. Compute $\frac{5}{6} + \frac{3}{8}$ and $\frac{5}{6} - \frac{1}{3}$.
3. Compute $\frac{3}{4} \div 6$ and $10 \div \frac{5}{7}$.
4. Compute $3\frac{1}{2} \div 1\frac{3}{4}$.
5. $\frac{4}{9}$ of a road is 12 km. Find the whole road.